

MAPPING PORTUGUESE MARKET STRUCTURE

*A Network Science Approach to
Portuguese Public Procurement*

Network Analysis

2025 / 26

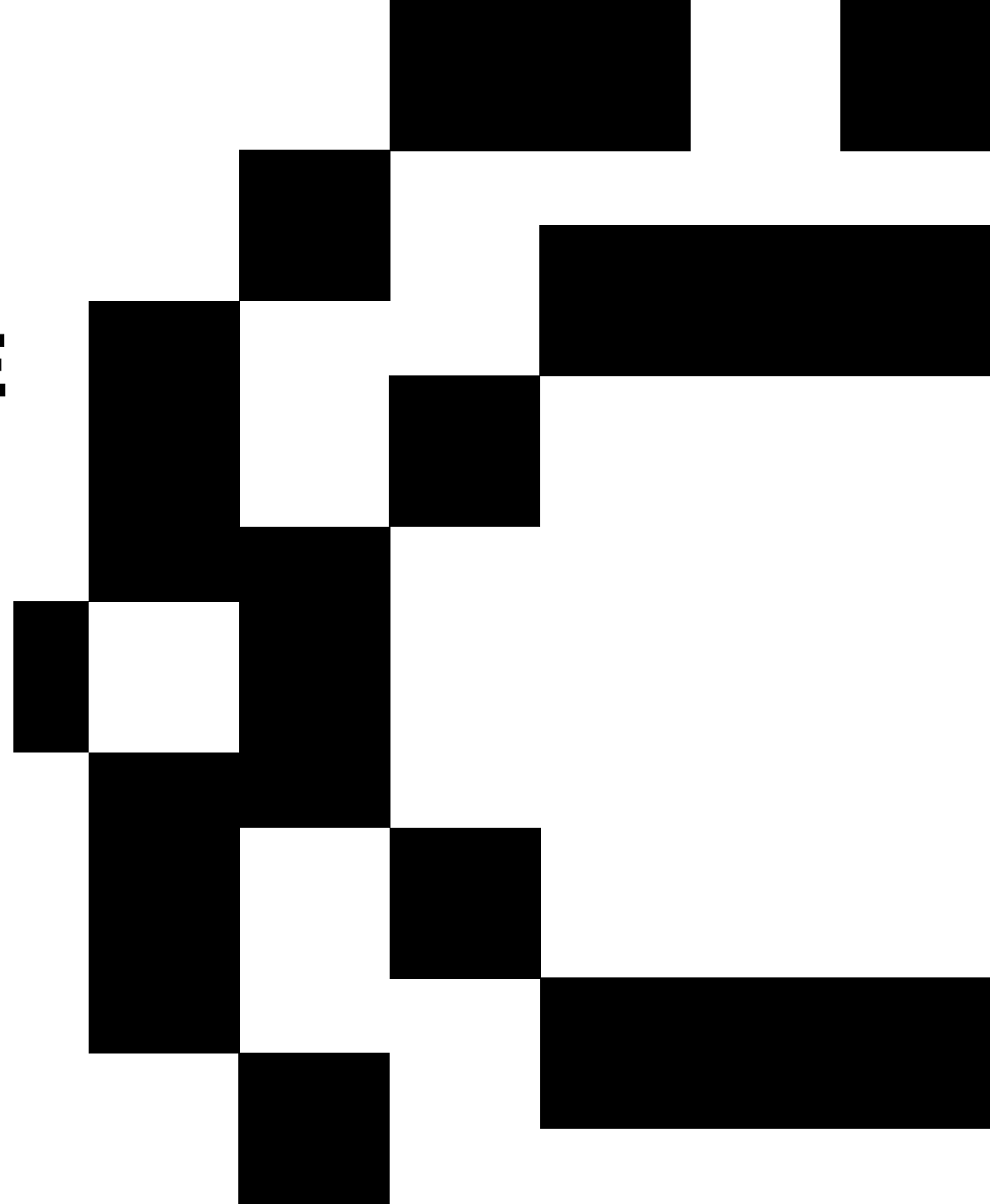
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3 QUESTIONS ABOUT €BILLIONS IN PUBLIC SPENDING

1

What does the market look like?

Macro-structure of the national procurement network

2

Who holds the power?

How concentrated is contract value among suppliers and buyers?

3

Is the capital different?

Does Lisbon amplify or soften the national patterns?

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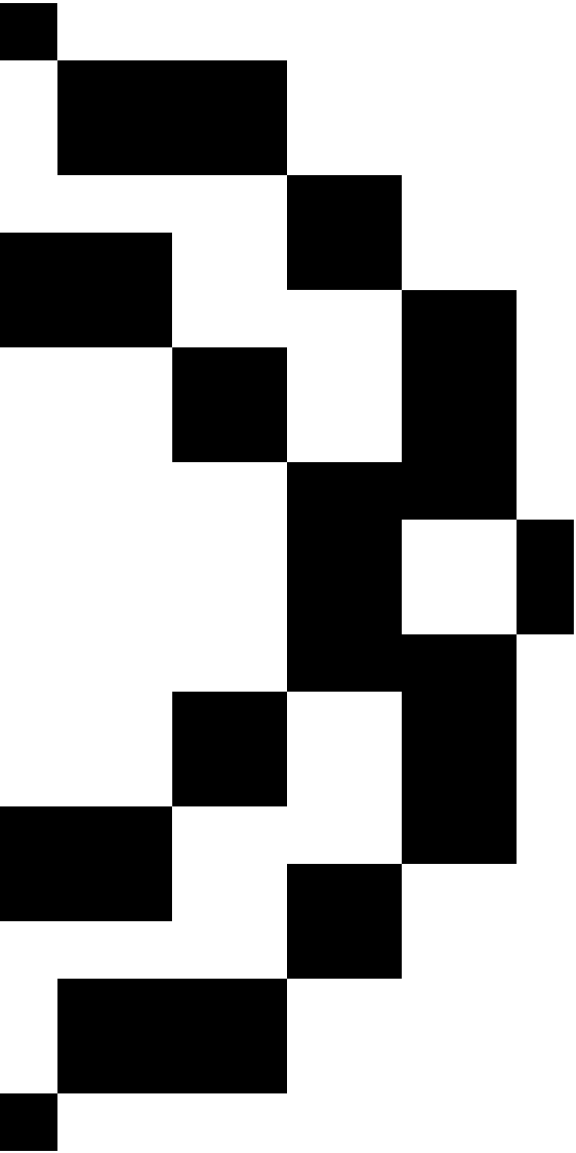
1 SOURCE & DATA

2 PREPROCESSING FUNNEL

3 NETWORK CONSTRUCTION

4 ZOOM IN

5 CONCLUSION



1

SOURCE & DATA

SOURCE & DATA

SOURCE

[Portal BASE](#) – Portuguese Public Procurement Portal

SUPPLEMENTARY

[dados.gov.pt](#) - Treated datasets

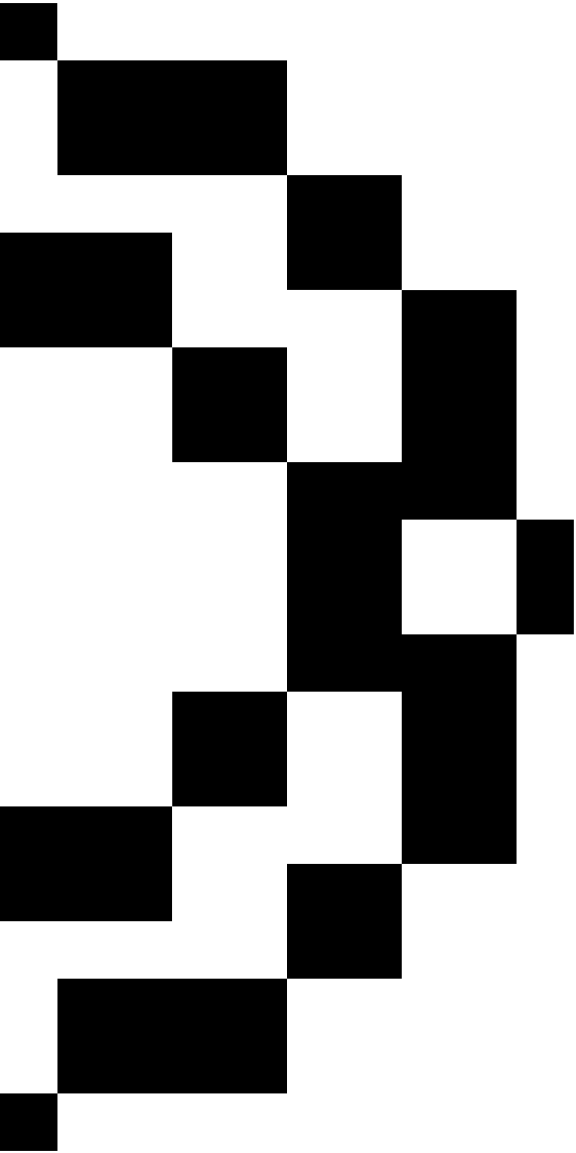
CONTRACT TIMELINE

Jan 1, 2023 → Apr 25, 2026

PROCEDURE TYPE

Public Procurement

CONTRACT VALUE	Nº COMPETITORS	CITY	CPV (agg)	PUBLIC ENTITY	COMPANY
118732.06	9	Ferreira do Alentejo	Construção	municipio de ferreira do alentejo	rdf construcoes sociedade imobiliaria
44709.43	12	Faro	Arquitetura e engenharia	direcao geral do livro dos arquivos e das bibliotecas	mota engil ativ gestao e manutencao e ativos
4389.00	19	Cascais	Saúde e equipamento médico	santa casa da misericordia de lisboa	boehringer ingelheim portugal

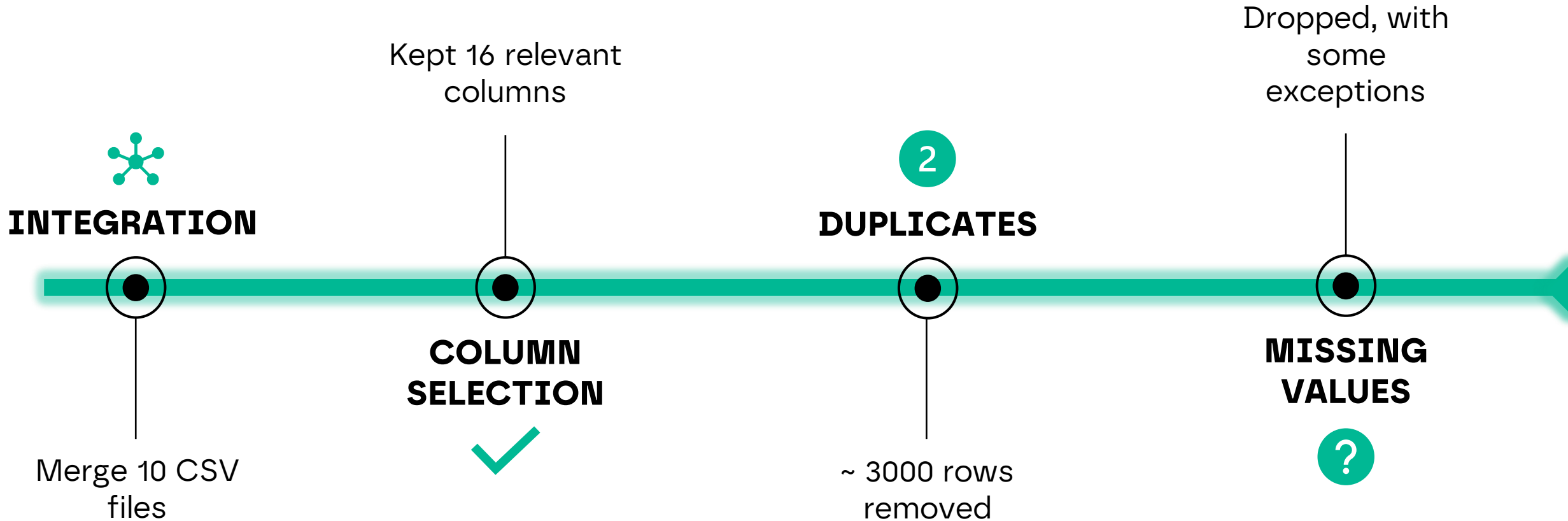


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PREPROCESSING FUNNEL

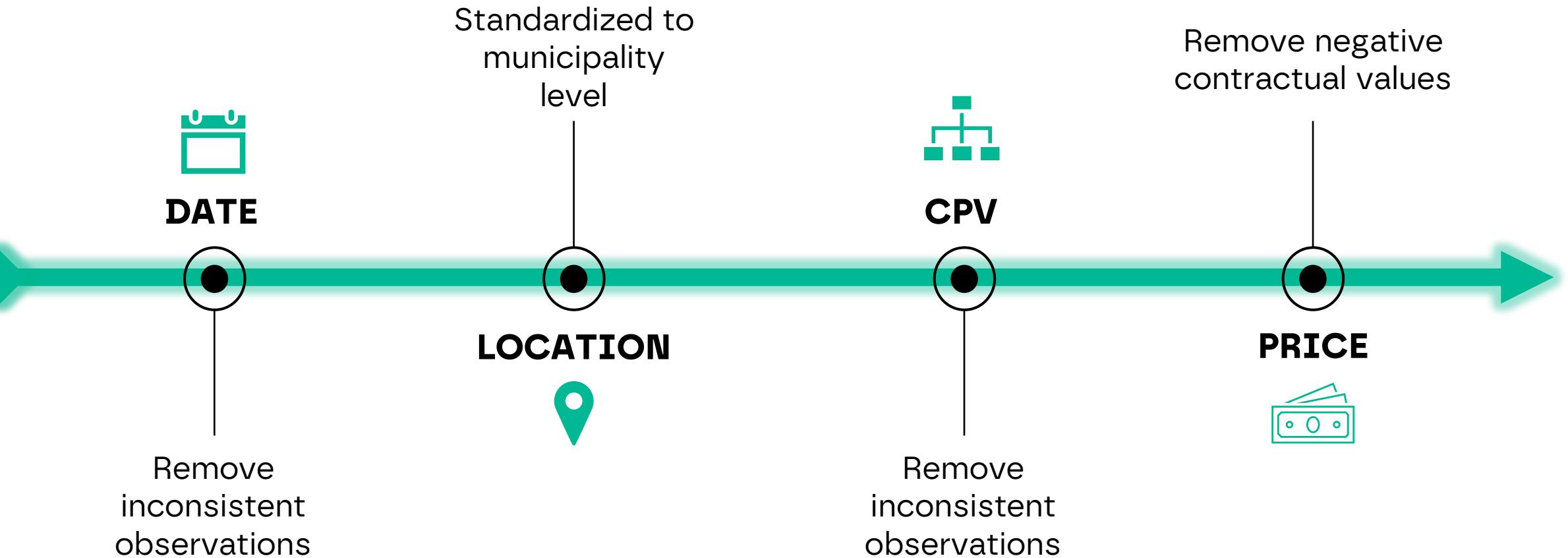
DATA PREPROCESSING

General Preprocessing



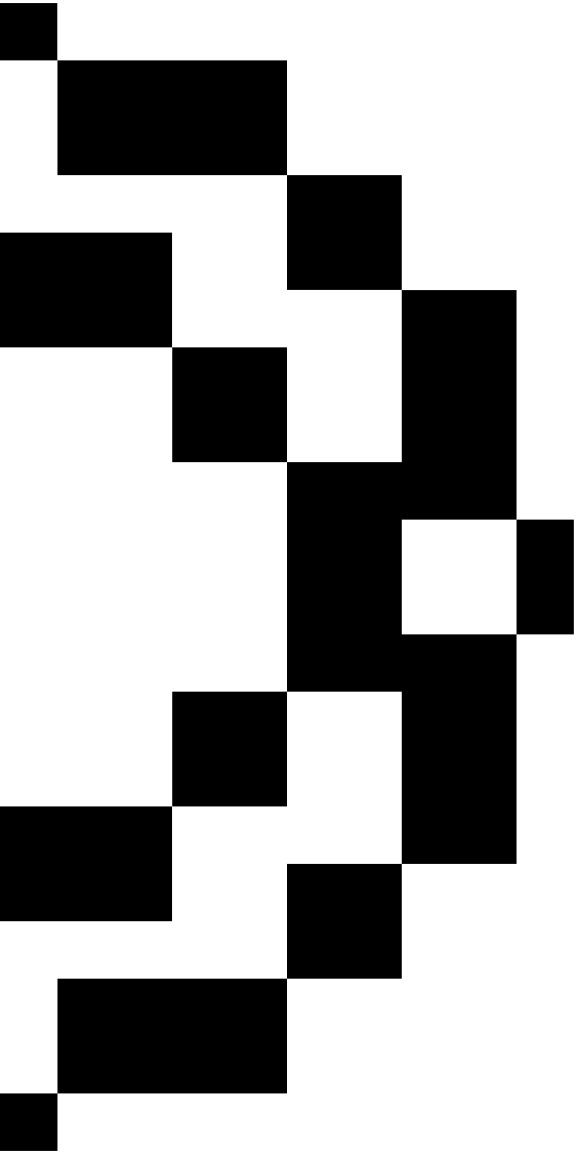
DATA PREPROCESSING

Specific Feature Preprocessing



PREPROCESSING EFFECT

	INITIAL STATUS	AFTER PREPROCESSING
Public Entities	10 309	1 109 (~10.8%)
Companies	143 783	5 338 (~3.8%)
Nodes	154 092	6 436 (~4.2%)
Links (Contracts)	732 573	16 989 (~2.3%)
Columns	35	22 (~62.8%)



3

NETWORK CONSTRUCTION

NETWORK DEFINITION

GRAPH TYPE

Bipartite · Directed · Weighted

BUYER NODES

Public Contracting Entities

SUPPLIER NODES

Private Companies / Contractors

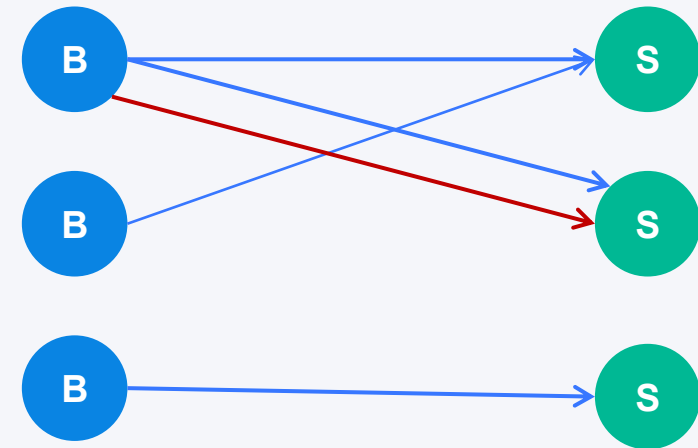
EDGES

Contractual relationships (Entity → Company)

EDGE WEIGHT

Contract value (precoContratual in €)

CONCEPTUAL SCHEMA



Buyer (Public Entity)



Supplier (Company)

NATIONAL PERSPECTIVE - PROPERTIES

HOW IS PUBLIC PROCUREMENT DISTRIBUTED?

Public procurement forms almost **one single interconnected marketplace**. **96%** of all 6,436 participants belong to the same connected ecosystem.

That connectedness is **thin, not dense**: the typical buyer works with only **~3-4 suppliers**, so participants tend to maintain a **limited number rather than broad** portfolios of **connections**.

The market is held together by **a minority of well-connected buyers** acting as meeting points. Any two players are only **~5 steps apart**.

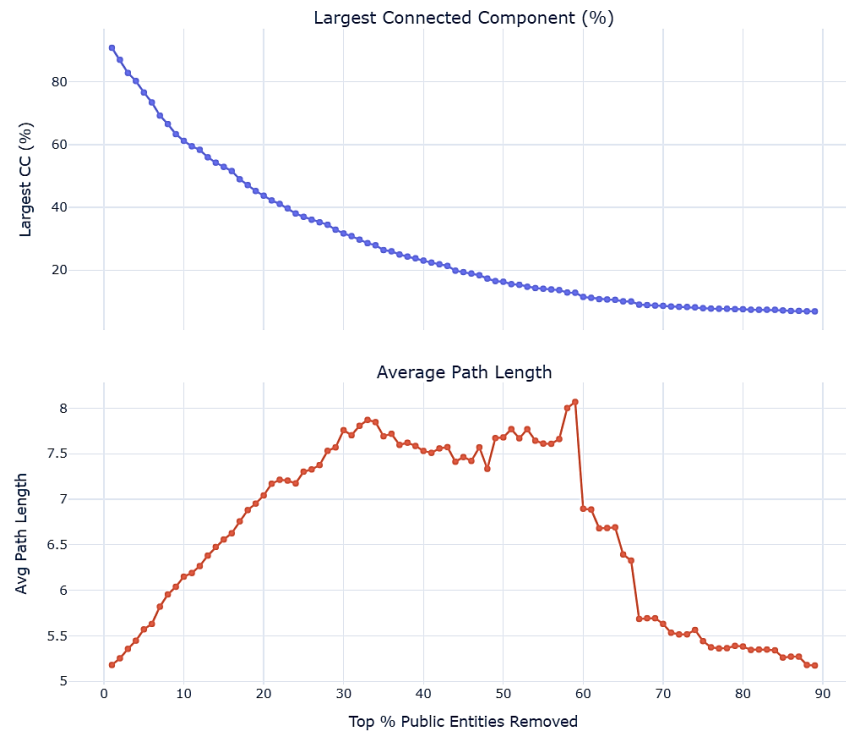
The **zero clustering** is not a weakness but the natural signature of a buyer → supplier market: two suppliers essentially never contract with each other.

METRIC	VALUE
Nodes (N)	6 436
Links (L)	10,995
Density	0.000265
Largest CC (%)	96.0%
Avg Degree (k)	3.42
Avg Strength (s) (€)	54
Avg Path Length (d)	5.07
Clustering Coefficient	0.000000
Transitivity	0.000020
Connected Components	113

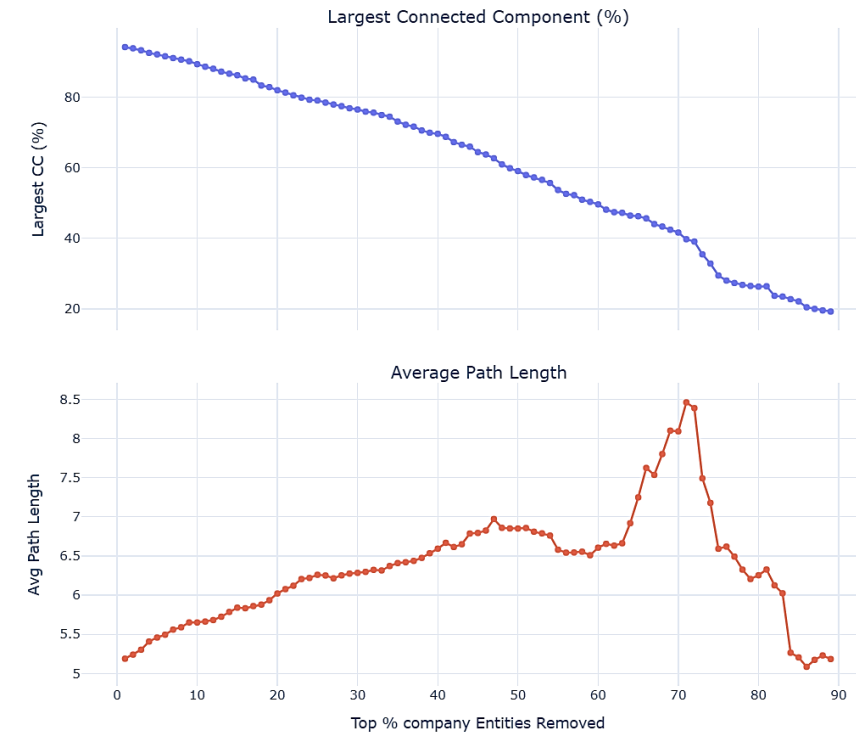
CONNECTIVITY & BRIDGES

HOW IS PUBLIC PROCUREMENT DISTRIBUTED?

PUBLIC ENTITIES



COMPANIES



From the 20 “bridges” (*betweenness*), 15 are Public Entities and only 5 are Private Companies.

Public entities are the major connectors of the network!

DEGREE DISTRIBUTION & POWER LAW

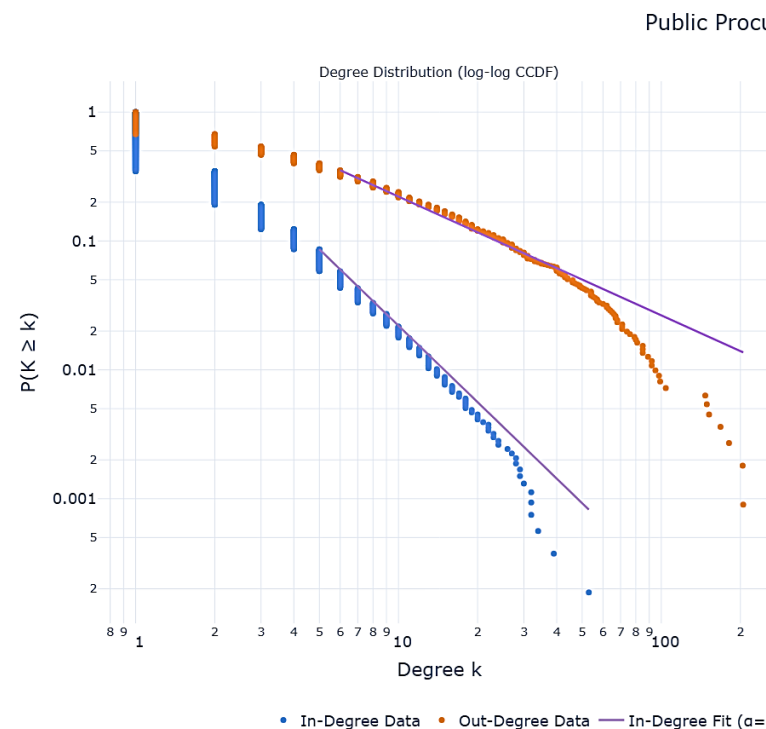
HOW IS PUBLIC PROCUREMENT DISTRIBUTED?

**Few connect to many;
most connect to few.**

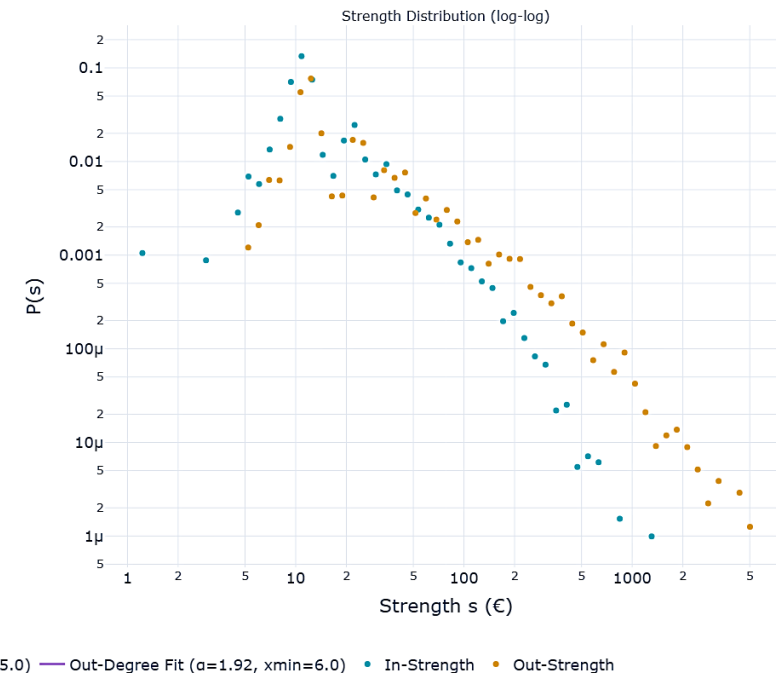
New entries struggle to
gain contracts.

**A "winner-takes-much"
market!**

Who Connects To Whom?



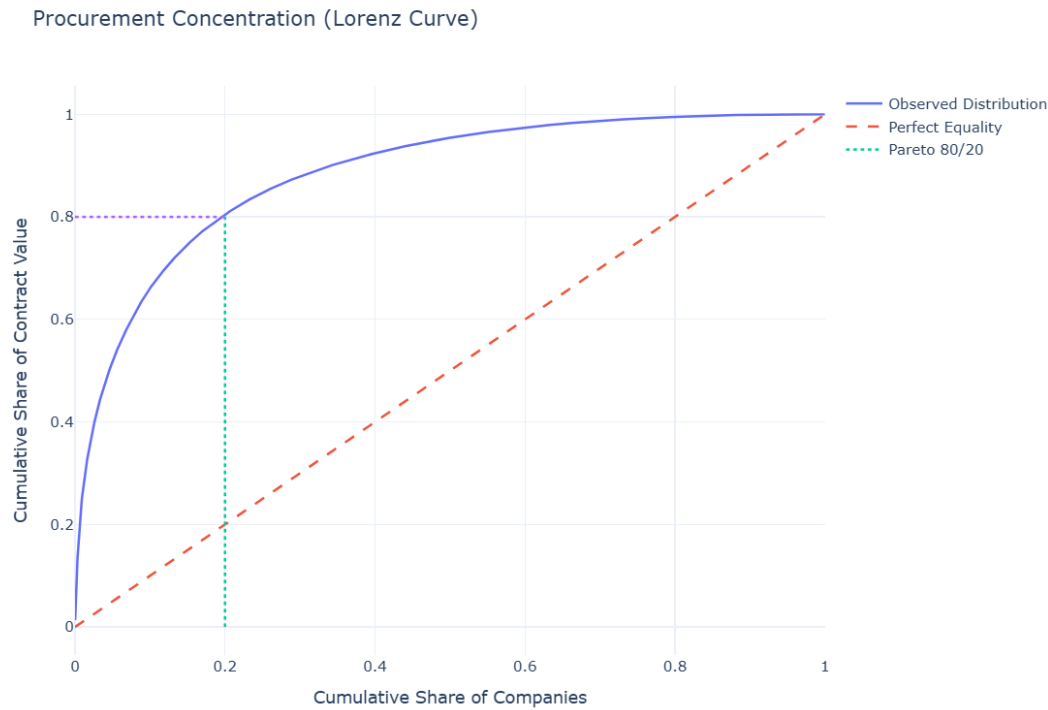
Where The Money Goes?



WEIGHTED DISTRIBUTION & VALUE CONCENTRATION

HOW HIGHLY CONCENTRATED IS THE NETWORK?

Companies & Public Entities



**Top 20% companies control
80% contract value**

Pareto Rule!

**Top 1% companies control
26% contract value**

Top end dominance

HHI coefficient: 20.65

No single company controls

DOMINANCE MEASUREMENT

No single number captures dominance. A player can be powerful in different ways, so we measure it through **5 complementary lenses**, applied to both.

LENS	QUESTION	COMPANY SIDE	PUBLIC ENTITY SIDE
Reach (Degree)	How many partners?	distinct public entities served	distinct companies used
Activity (Contracts)	How many deals?	contracts won	contracts issued
Money (Total Value)	How many euros?	€ captured	€ spent
Influence (PageRank)	How central to important players?	sought by big public entities	feeds big companies
Gatekeeping (Betweenness)	How much it bridges the market?	links separate sectors	links separate supplier groups

True Dominance = scoring high on several lenses at once, not just one.

NATIONAL PERSPECTIVE

WHICH ARE THE DOMINANT COMPANIES?

COMPANY	TOP-10 APPEARANCES	IN- DEGREE	ACTIVITY	MONEY
<i>MEO</i>	4 / 5	30	50	28.1M
<i>Claranet II Solutions</i>	4 / 5	53	118	16.9M
<i>Petrogal</i>	3 / 5	29	51	14.7M
<i>Sogenave</i>	2 / 5	32	152	8.8M
<i>Gertal</i>	2 / 5	22	34	16.7M
<i>Base2</i>	2 / 5	39	52	6.9M
<i>Exumas Consulting</i>	2 / 5	34	87	2.1M
<i>Nautilus</i>	2 / 5	32	41	2.0M

NATIONAL PERSPECTIVE

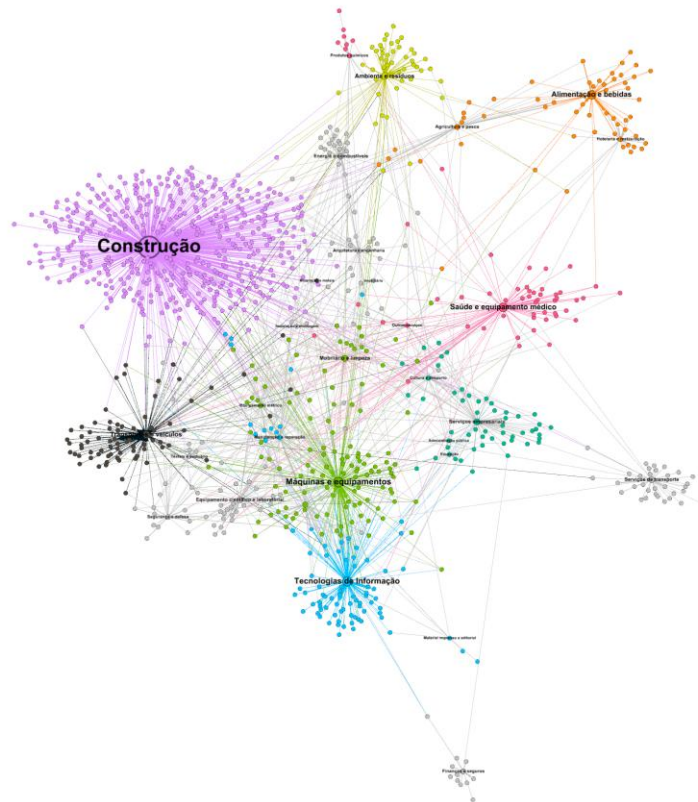
WHICH ARE THE DOMINANT PUBLIC ENTITIES?

PUBLIC ENTITY	TOP-10 APPEARANCES	OUT- DEGREE	ACTIVITY	MONEY
INSA (Inst. Nac. Saúde Ricardo Jorge)	4 / 5	85	227	3.4M
<i>Guarda Nacional Republicana</i>	<i>3 / 5</i>	<i>204</i>	<i>554</i>	<i>27.3M</i>
<i>Universidade do Minho</i>	<i>3 / 5</i>	<i>38</i>	<i>43</i>	<i>3.3M</i>
<i>ISEP (Eng. Porto)</i>	<i>3 / 5</i>	<i>47</i>	<i>86</i>	<i>1.1M</i>
<i>Universidade de Aveiro</i>	<i>3 / 5</i>	<i>30</i>	<i>34</i>	<i>3.3M</i>
<i>ISEG (Economia & Gestão)</i>	<i>3 / 5</i>	<i>46</i>	<i>49</i>	<i>3.1M</i>
<i>Fundação INATEL</i>	<i>3 / 5</i>	<i>28</i>	<i>44</i>	<i>2.5M</i>
Infraestruturas de Portugal	2 / 5	98	167	92.7M

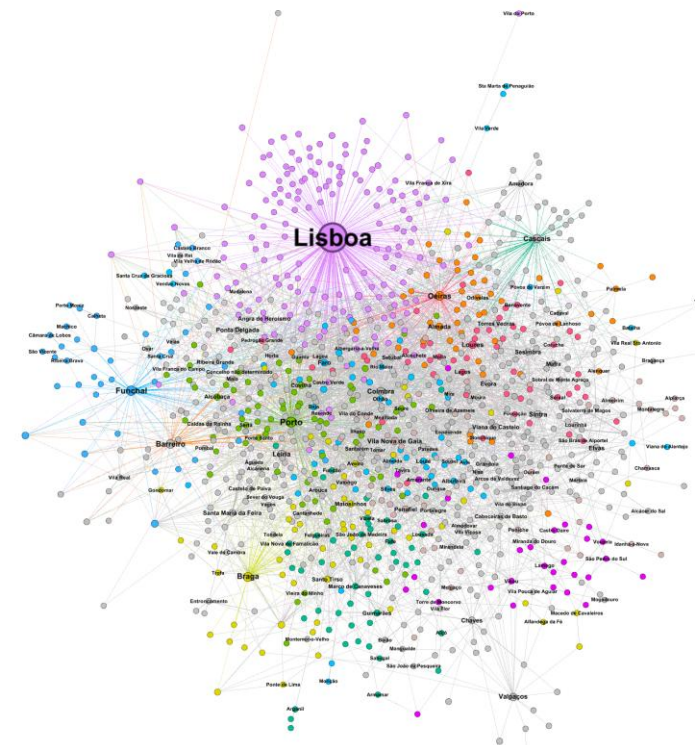
NATIONAL PERSPECTIVE

WHY ARE COMPANIES SPECIALIZED?

CPV



CITY



COMMUNITY DETECTION

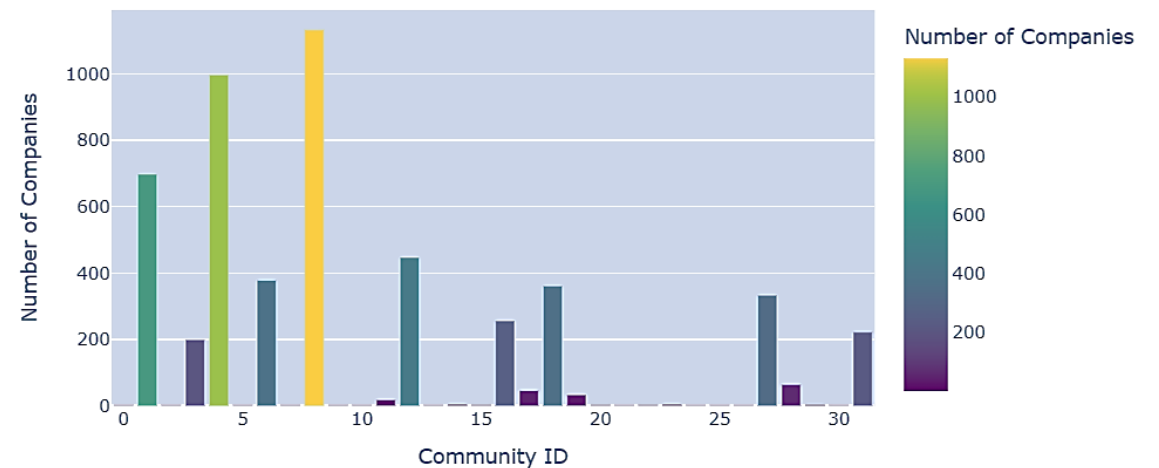
The market splits naturally into **32 distinct segments**, organised primarily by sector and secondarily by geography, with Lisbon recurring as the dominant location.

The segmentation is **strong and real** (modularity $Q \approx 0.55 > 0.3 \rightarrow$ significant community structure).

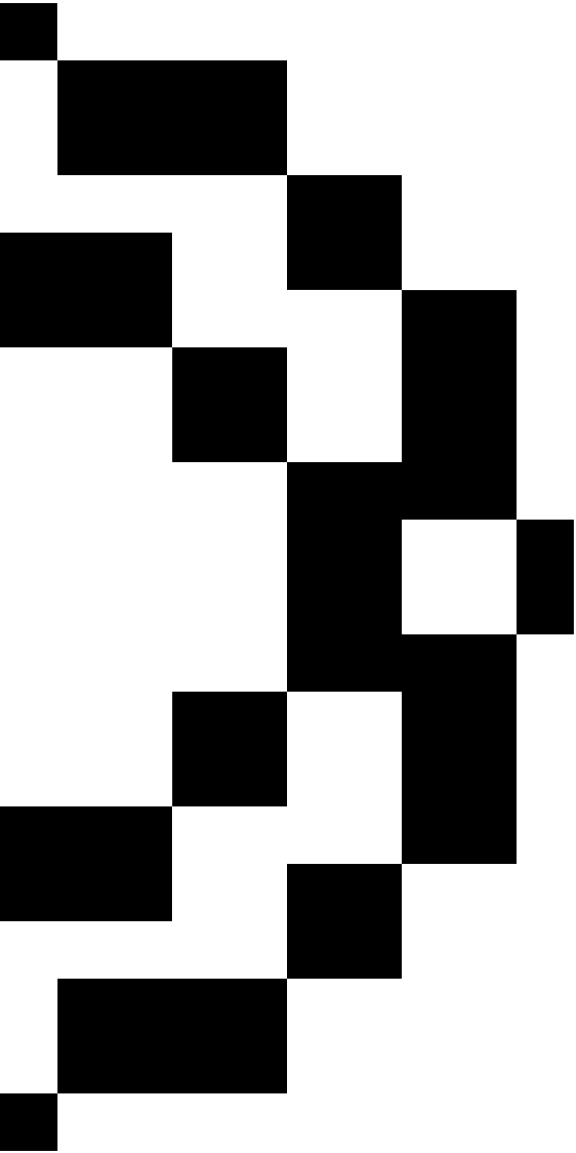
Companies that co-supply the same buyers form **very tight groups**: once companies share a public client, they tend to cluster densely.

The **most valuable** segment is the **IT/software** cluster led by Claranet II Solutions (~€16.8M). It also had the most of the key bridge companies.

Community Size Distribution (Louvain, $Q=0.553$)



Community size statistics:
Largest community: 1133 companies
Smallest community: 2 companies
Mean size: 163.5
Median size: 4.5

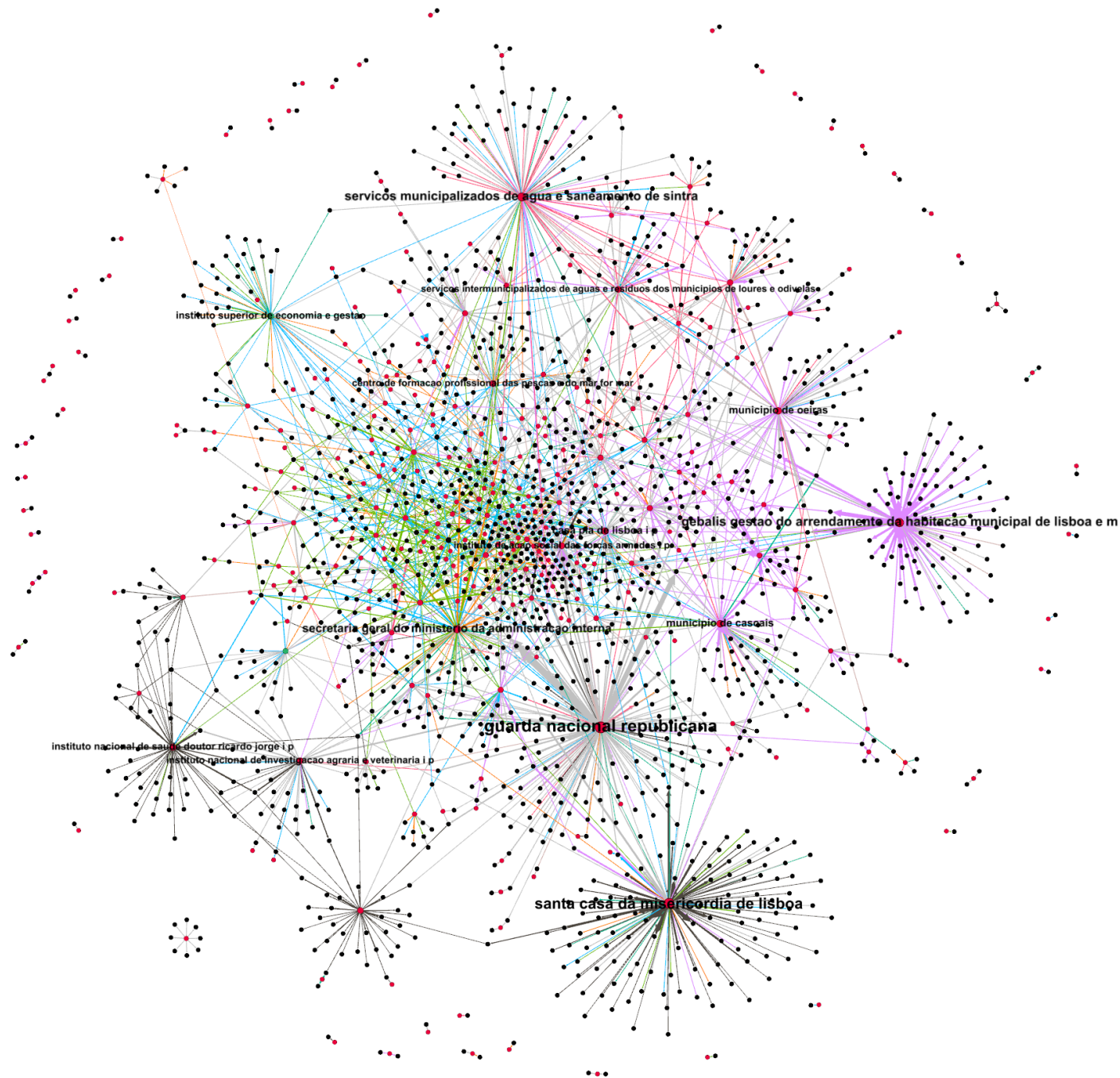


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ZOOM IN PORTUGAL → LISBON

LISBON VS PORTUGAL

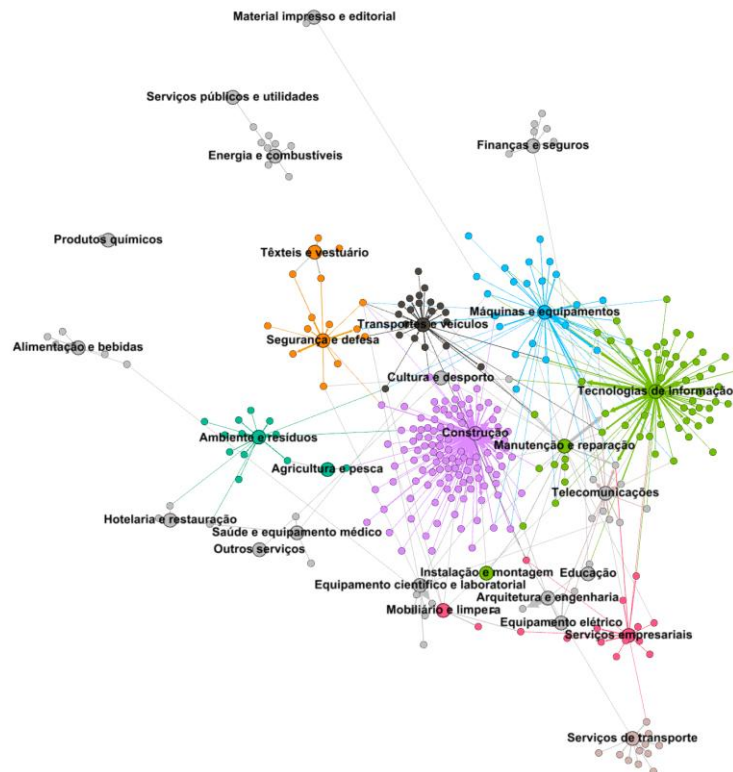
METRIC	LISBON	PORTUGAL	READING
Nodes (N)	1 824	6 436	~28% of players
Links (L)	2 432	10 995	~22%
Density	0.000731	0.000265	Lisbon x2.8 denser
Avg Degree (k)	2.67	3.42	slightly lower
Avg Path Length (d)	4.97	5.07	almost identical
Largest CC (%)	93.8%	96.0%	one giant market both



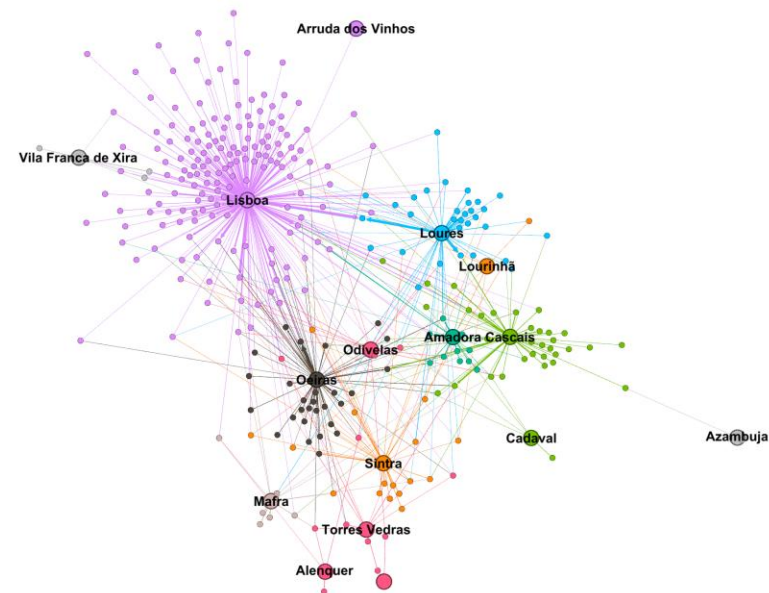
LISBON PERSPECTIVE

WHY ARE COMPANIES SPECIALIZED?

CPV



CITY



CONCENTRATION

Lisbon concentrates demand and strategic IT influence more strongly than Portugal overall.

	PORTUGAL	LISBON
Companies concentration (Gini)	0.769	0.742 ↓
Top 20% companies share	80.5%	77.6% ↓
Public Entities concentration (Gini)	0.750	0.809 ↑
Top 20% public entities share	78.6%	85.5% ↑

- **Claranet and MEO dominate both networks**, with 52% of Claranet's national value originating in Lisbon.
- **Nationally**, contract value is concentrated in **telecoms, transport, and energy**.
- In **Lisbon**, **IT/software firms** occupy the **most central** positions, with **construction and medical equipment** account for **large contract volumes**.

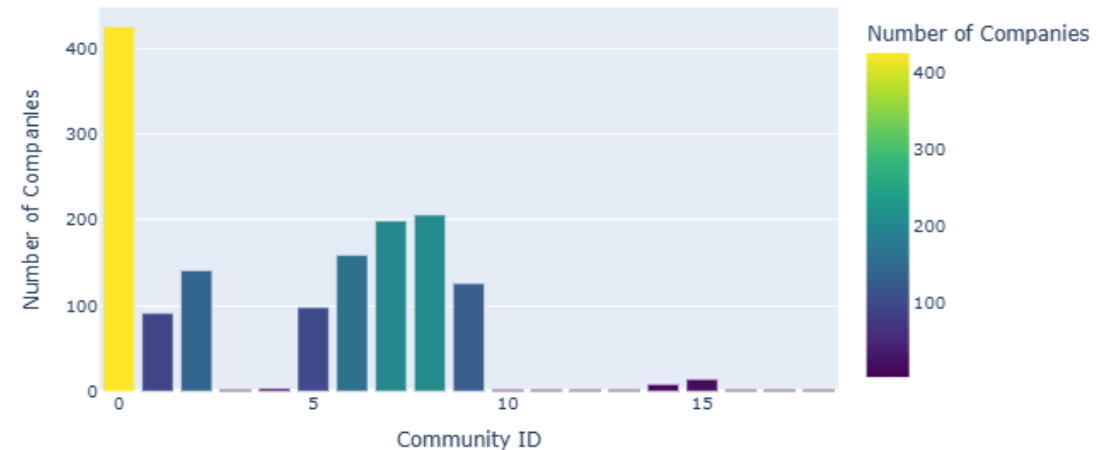
COMMUNITY STRUCTURE

The Lisbon market is structured into **19 clear, dense communities**, with a strong concentration in **tech and services**.

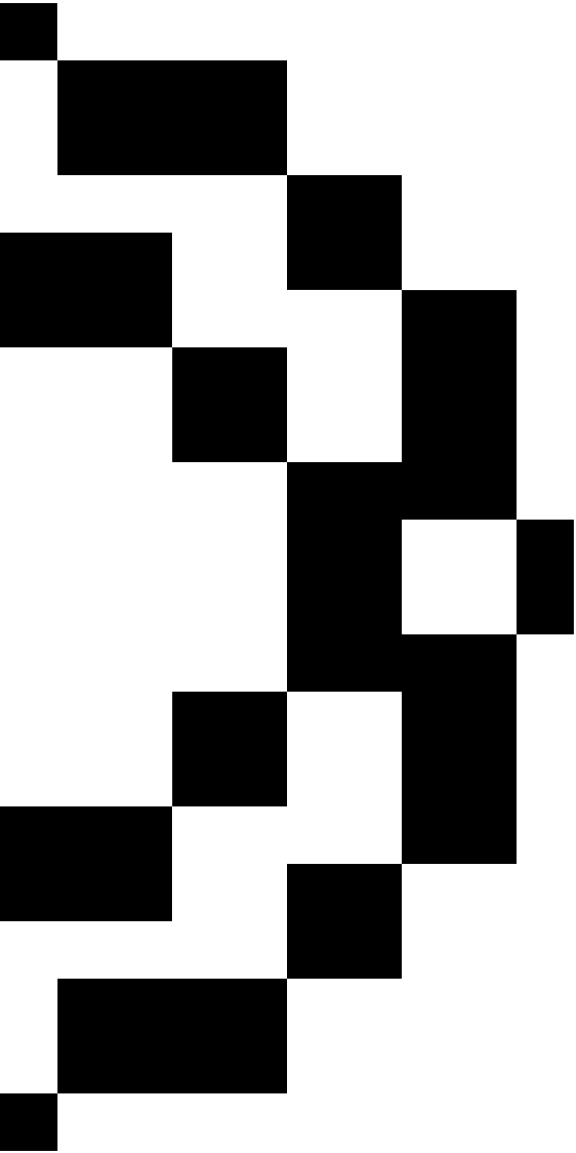
The network shows robust **modularity ($Q \approx 0.55$)**, confirming well-defined community structure. Companies that share clients tend to co-locate within the same tight clusters, reinforcing localised supply chains.

The **IT/software cluster dominates** the landscape, led by **Claranet II Solutions** (~€16.8M), also acting as a key connector in the network.

Community Size Distribution (Louvain, $Q=0.673$)



Community size statistics:
Largest community: 426 companies
Smallest community: 2 companies
Mean size: 78.3
Median size: 8.0



5

CONCLUSION

- What does the market look like?*
A single small-world, hub-dominated marketplace.
- Who holds the power?*
IT champions (Claranet, MEO) are structural gatekeepers.
- Is the capital different?*
Lisbon = denser buyers + dominant IT + sharper communities

Lisbon doesn't just mirror Portugal, it concentrates **demand** and **crowns IT** as the connective tissue of public spending.

The market is structured around **public entities acting as hubs**, with **companies clustered** according to their **area of business** (CPV).

Thank You!

GITHUB REPOSITORY: <https://github.com/beatrizmarques2005/network-analysis>

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